INFOSYS SP & DSE – BATCH 4

Week 1 practice sheet: Dynamic programming

Solve in order. Aim for 45 minutes per hard problem before checking the hints.

#	PROBLEM	LEVEL	PATTERN
1	Climbing stairs	Easy	1-D recurrence
2	0/1 knapsack	Medium	Take / skip table
3	Coin change (minimum coins)	Medium	Unbounded knapsack
4	Longest increasing subsequence	Hard	DP + binary search
5	Edit distance	Hard	2-D table on prefixes
6	Partition equal subset sum	Medium	Subset-sum bitset

Hints

1. Write the recursive solution first, then list which arguments change. Those become the table's dimensions.
2. Coin change: *dp[a]* = 1 + min over coins *c* of *dp[a − c]*, with *dp[0]* = 0. Fill amounts from 0 upwards.
3. LIS in O(n log n): keep *tails[k]*, the smallest tail of an increasing subsequence of length k + 1, and binary-search where each number goes.
4. Edit distance: compare the last characters of both prefixes; three moves cover insert, delete and replace.

Before Friday's mock

- Submit problems 2–4 on the practice portal and note your time for each.
- Bring one problem you could not finish; we will debug two of them live.

Sample material created for the UnsaidTalks demo.